

Aim / Objective

To find the longest substring without repeating characters using Java string methods, loops, and Scanner input.

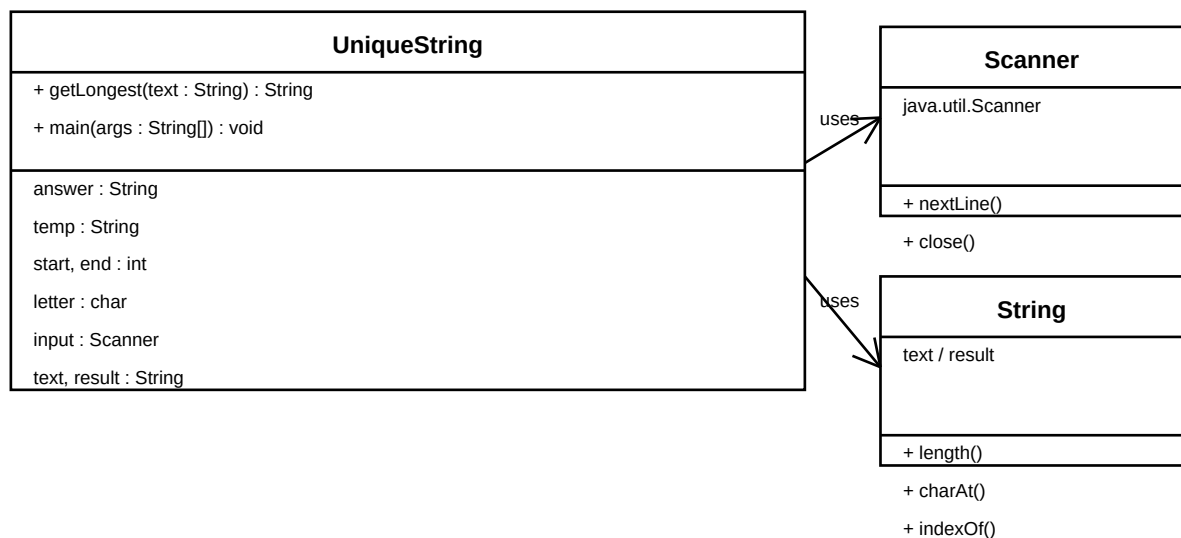
System Requirements

Java Development Kit (JDK) and any Java IDE such as Eclipse or IntelliJ IDEA.

Algorithm / Logic Description

1. Read a string using Scanner.
2. Start from each character and create a temporary substring.
3. Add characters while checking for duplicates using indexOf().
4. Stop when a duplicate character is found.
5. Compare the current substring length with the longest result.
6. Display the longest length and substring.

UML Class Diagram



Source Code

```
package core_java;

import java.util.Scanner;

public class UniqueString {

    static String getLongest(String text) {

        String answer = "";

        for (int start = 0; start < text.length(); start++) {

            String temp = "";

            for (int end = start; end < text.length(); end++) {

                char letter = text.charAt(end);
                // Check for duplicate character
```

```

        if (temp.indexOf(letter) >= 0) {
            break;
        }

        temp += letter;

        // Compare current substring with previous result
        if (temp.length() >= answer.length()) {
            answer = temp;
        }
    }
}

return answer;
}

public static void main(String[] args) {

    Scanner input = new Scanner(System.in);

    System.out.print("Enter a string: ");
    String text = input.nextLine();

    String result = getLongest(text);

    System.out.println("Longest length: " + result.length());
    System.out.println("Longest substring: " + result);

    input.close();
}
}

```

Compilation and Execution Commands

```

javac UniqueString.java
java UniqueString

```

Output

```

PS C:\Users\Student> javac UniqueString.java
PS C:\Users\Student> java UniqueString
Enter a string: abcabcbb
Longest length: 3
Longest substring: abc
PS C:\Users\Student>

```

Result

The program was implemented successfully to find the longest substring containing unique characters.